

FPV CAMERAS FOR DRONES: An Introduction



T.K. Hareendran
is founder
and promoter
of TechNode
Pratolabz—an
electronics R&D Lab

What pilots see while flying drones is the low-latency analogue video from a first-person view (FPV) camera. FPV camera is a compact and lightweight module mounted on the drone that broadcasts real-time video back down to the pilot with the help of a dedicated video transmitter module. Here are the things to consider when equipping your drone with an FPV camera.

FPV system overview

First-person view, also called video piloting, means flying the drone from the perspective of a true pilot by using an onboard video camera and wireless downlink. In the last couple of years, various vendors have started to sell kits that contain the minimum FPV system components needed to outfit a drone for FPV flight. First, let us go through the various components that you need to upgrade/build a drone for FPV use.

Video camera. FPV cameras use either an inexpensive complementary

metal-oxide-semiconductor (CMOS) or a bit pricey charge-coupled device (CCD) image sensor. The image sensor of the camera fundamentally affects its performance. Also note that a camera with a wide input voltage range offers flexibility when wiring the FPV system.

Video transmitter. Video transmitter takes the analogue signal from the camera and sends it out through an omnidirectional antenna. As the video transmitter is typically designed to work in one of several frequency bands, usually a channel selector switch is provided on the transmitter to set the exact frequency for video signal transmission.

Video receiver. Video receiver with an appropriate antenna is a vital component of the ground station. Output of the video receiver is fed to a portable video display (small-screen video monitor, or video goggles). Usually, standard lithium polymer (Li-Po) batteries are used as the power source for the video receiver and monitor in all ground stations.

FPV camera insights

Image sensor. The image quality of CCD cameras is undoubtedly better than CMOS. However, CMOS cameras are cheaper than CCD, and provide good enough performance for most FPV applications.

Latency. The time between a stimulus and the response is called latency. It reflects the processing speed of the camera. Faster processing lowers the latency, and an FPV camera with a latency of less than 40ms is a standard choice.

Field of view. Expressed as an angle, it is a measure of how much the camera can see around. Usually, for FPV applications, having a wider field of view is better as it allows you to see more around you. However, this can sometimes show distorted images with curved horizons.

TVL resolution. TV line (TVL) is a

Fig. 1: FPV drone

